

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

A LIGA Fabricated Two-Dimensional Quadrupole Array for High Resolution Mass Spectroscopy

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ABSTRACT

Microfabrication of high precision two-dimensional quadrupole array (3 X 3) is currently in progress at JPL using the LIGA processing. The target detection range for the miniaturized mass spectrometer is 1 to 300 atomic mass units (AMU) with a resolution of 0.5 AMU. Mass spectrometer is one of most frequently flown instruments for planetary aeronomy studies. Development of small and portable high precision in-situ instrument are greatly needed for human environmental monitoring, geochronology, and atmospheric characterization, which is also coincide with NASA's goal of having "faster, better, cheaper" space mission. LIGA processing is combination of x-ray lithography using synchrotron radiation, electroforming (electrodeposition), and plastic molding to fabricate a high precision device. It contains two distinct electrodeposition steps. One is the electrodeposition of gold onto a mask substrate to serve as the x-ray absorber during synchrotron exposure and the other is the deposition of actual device. By controlling electrodeposition parameters and solution compositions, homogeneous, low-stressed, and dense gold mask (20 to 50 micron) was electrodeposited from gold sulfite baths to silicon substrate (100-200 micron) to provide an adequate contrast ratio. 3 mm tall polymethyl methacrylate (PMMA) was precisely patterned and developed by controlling the filter thickness, energy dosage and time. Then copper was electrodeposited to PMMA mold from acid sulfate bath. Metrology studies were also conducted to determine the dimensional precision of the quadrupoles.